Radxa CM4 on DF40C sockets

Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

J9

GND	2	1	GND
	Ethernet_Pair1_P	Ethernet_Pair3_P	
	Ethernet_Pair1_N	Ethernet_Pair3_N	
GND	6	7	GND
	Ethernet_Pair0_N	Ethernet_Pair2_N	
	Ethernet_Pair0_P	Ethernet_Pair2_P	
GND	10	11	GND
	Ethernet_SYNC_IN(1.8v)	Ethernet_nLED3(3.3v)	
	Ethernet_SYNC_OUT(1.8v)	Ethernet_nLED2(3.3v)	
	EEPROM_nWP	Ethernet_nLED1(3.3v)	
GND	14	15	
	GPIO26	PI_nLED_Activity	
SAI2_LRCK	16	17	GND
SPI2_MOSI	GPIO19	GPIO21	SAI2_SDO
SPI_INT	GPIO13	GPIO20	SAI2_SDI
	GPIO6	GPIO16	
GND	20	21	SPI2_CLK
SPI_IO	GPIO5	GPIO12	
CAM_SDA	22	23	GND
	ID_SD	ID_SC	CAM_SCL
	GPIO11	GPIO7	SERVO_PWM
	GPIO9	GPIO8	
GND	26	27	
EYE_DATA	GPIO10	GPIO25	GND
SPI2_MISO	28	29	
	GPIO22	GPIO24	UART7_RX
	GPIO27	GPIO23	UART7_TX
FAN_PWM	GPIO17	GPIO18	SAI2_SCLK
GND	32	33	
SPI2_CS	GPIO4	GPIO14	GND
I2C8_SCL	GPIO3	GPIO15	
I2C8_SDA	GPIO2	SD_CLK	
GND	38	39	GND
	SD_CMD	SD_DAT3	
	SD_DAT5	SD_DAT0	
GND	42	43	GND
	SD_DAT4	SD_DAT1	
	SD_DAT7	SD_DAT2	
	SD_DAT6	47	GND
GND	48	SD_VDD_Override	
	Reserved	SD_PWR_ON	
CM4_3V3	GPIO_VREF(1.8v/3.3v_Input)	+5v_(Input)	+5V
	SCL0	+5v_(Input)	+5V
	SDA0	+5v_(Input)	+5V
CM4_3V3	+3.3v_(Output) 600mA Max	+5v_(Input)	+5V
	+3.3v_(Output)	+5v_(Input)	+5V
	+1.8v_(Output) 600mA Max	+5v_(Input)	+5V
	+1.8v_(Output)	WiFi_nDisable	
CM4_RESET	60	61	
	RUN_PG	BT_nDisable	
	AnalogIP1	nRPIBOOT	CM4_MASKROM
	AnalogIP0	nPI_LED_PWR	
GND	66	67	CAM_GPIO
	nEXTRST	Camera_GPIO	
		Global_EN	

NB SD signals are only available on modules without eMMC

CM4 socket 1–100

J10

	102	101	
	PCIe_CLK_nREQ	USB_OTG_ID	
	104	103	USB_DN
	Reserved	USB2_N	USB_DP
	106	105	
GND	Reserved	USB2_P	
	108	107	GND
	GND	GND	
	110	109	
	PCIe_CLK_P	PCIe_nRST	
	112	111	
	PCIe_CLK_N	VDAC_COMP	
GND	114	113	
	GND	GND	GND
	116	115	
	PCIe_RX_P	CAM1_D0_N	
	118	117	
	PCIe_RX_N	CAM1_D0_P	
GND	120	119	GND
	GND	GND	
	122	121	
	PCIe_TX_P	CAM1_D1_N	
	124	123	
	PCIe_TX_N	CAM1_D1_P	
GND	126	125	
CSI_D0_N	GND	GND	GND
CSI_D0_P	128	127	
	CAM0_D0_N	CAM1_C_N	
	130	129	
	CAM0_D0_P	CAM1_C_P	
GND	132	131	GND
CSI_D1_N	GND	GND	
CSI_D1_P	134	133	
	CAM0_D1_N	CAM1_D2_N	
	136	135	
	CAM0_D1_P	CAM1_D2_P	
GND	138	137	GND
CSI_C_N	GND	GND	
CSI_C_P	140	139	
	CAM0_C_N	CAM1_D3_N	
	142	141	
	CAM0_C_P	CAM1_D3_P	
GND	144	143	
	GND	HDMI1_HOTPLUG	
	146	145	
	HDMI1_TX2_P	HDMI1_SDA	
	148	147	
	HDMI1_TX2_N	HDMI1_SCL	
GND	150	149	GND
	GND	HDMI1_CEC	
	152	151	
	HDMI1_TX1_P	HDMI0_CEC	
	154	153	
	HDMI1_TX1_N	HDMI0_HOTPLUG	
GND	156	155	GND
	GND	GND	
	158	157	
	HDMI1_TX0_P	DSI0_D0_N	
	160	159	
	HDMI1_TX0_N	DSI0_D0_P	
GND	162	161	GND
	GND	GND	
	164	163	
	HDMI1_CLK_P	DSI0_D1_N	
	166	165	
	HDMI1_CLK_N	DSI0_D1_P	
GND	168	167	GND
	GND	GND	
	170	169	
	HDMI0_TX2_P	DSI0_C_N	
	172	171	
	HDMI0_TX2_N	DSI0_C_P	
GND	174	173	GND
	GND	GND	
	176	175	
	HDMI0_TX1_P	DSI1_D0_N	
	178	177	
	HDMI0_TX1_N	DSI1_D0_P	
GND	180	179	GND
	GND	GND	
	182	181	
	HDMI0_TX0_P	DSI1_D1_N	
	184	183	
	HDMI0_TX0_N	DSI1_D1_P	
GND	186	185	GND
	GND	GND	
	188	187	
	HDMI0_CLK_P	DSI1_C_N	
	190	189	
	HDMI0_CLK_N	DSI1_C_P	
GND	192	191	GND
	GND	GND	
	194	193	
	DSI1_D3_N	DSI1_D2_N	
	196	195	
	DSI1_D3_P	DSI1_D2_P	
GND	198	197	GND
	GND	GND	
	200	199	
	HDMI0_SCL	HDMI0_SDA	

High Speed Serial

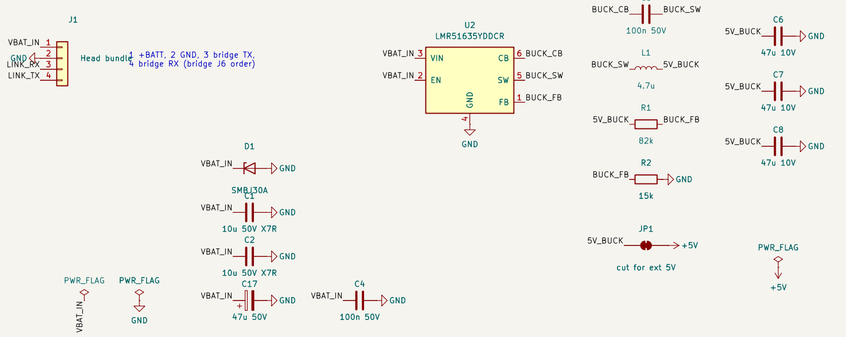
CM4 socket 101–200

GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

H5 — Radxa CM4

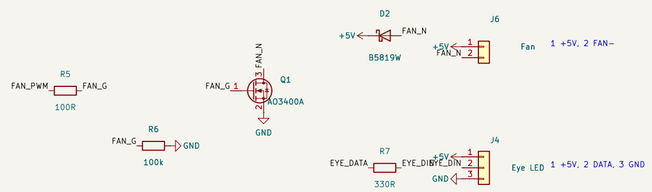
Head bundle in, 5 V buck

LMR51635Y, 1.1 MHz, 4.7 uH (Ti table 8-2: 3.3 uH), 82k/15k sets 5.17 V. EN tied to VIN. F6 (2 A) on the bridge fuses this feed.
JP1 joins the buck to the 5 V rail: cut it to feed 5 V from outside (J12). The feedback divider stays on the buck side



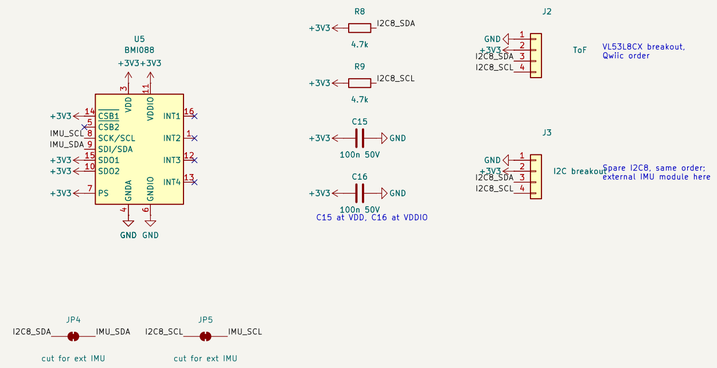
Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data



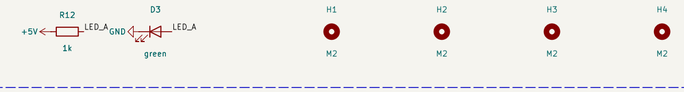
I2C8: head IMU, ToF, breakout

BMI088 on the board. I2C: PS and CSB1 high, CSB2 open (allowed in I2C mode): SDO1 and SDO2 high: accelerometer 0x19, gyroscope 0x69 (strapped so the pin rows need no vias between alternate pins).
R8, R9: the bus pull-ups (the ToF breakout adds 10k). Cut JP4 and JP5 to use an external IMU module on J3



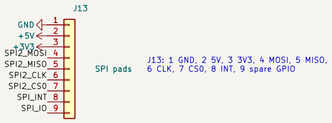
Power LED, mounting holes

M2 holes, one per corner, not connected



SPI breakout

SPI2 M1 for a future SPI device, untested. Wire pads on the bottom, under the module, along the top edge.
CLK GPIO12 (CM4 31), MOSI GPIO13 (28), MISO GPIO22 (46), CS0 GPIO4 (54), INT GPIO6 (30), spare GPIO5 (34)



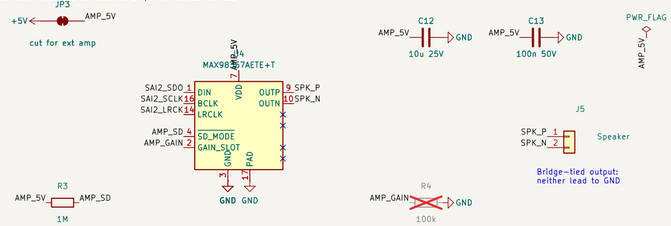
3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout



Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono. GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB.
JP3 feeds it: cut JP3 to use an external amplifier module on J11

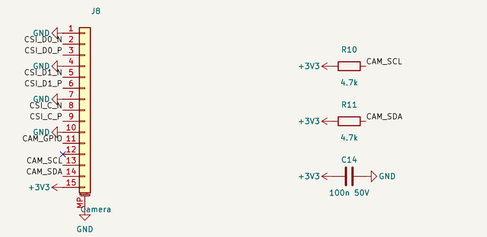


Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)

Camera FPC

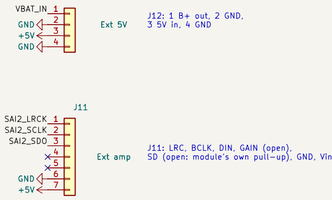
Pi 15-pin camera order. CAM0 lanes (Radxa CSI1), I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs



Pin 12 (camera clock) not connected: the module has its own oscillator

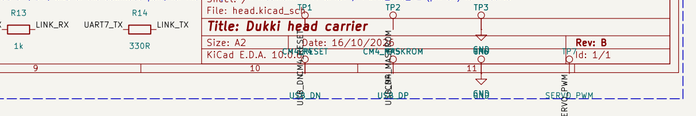
External fallbacks

Wire pads, 2.54 mm. J12: cut JP1 and feed 5 V from an external step-down. J11: cut JP3 and wire an amplifier module, pins in the generic module's order. J3 (section 6) takes an external IMU module



Link series resistors, test pads

Series R in each UART7 line: 1k on RX limits current into an unpowered CM4 from the bridge, 330R on TX. TP1/TP2 (bottom): short to TP3 to reset; hold TP2 low through a reset for maskrom. TP4–TP6 (bottom): USB D−, D+, GND of the download port, for rkdeveloptool. TP7: GPIO7 / PWM0_CH0_M1 (pin 37) for a servo



Bootstrapped by scripts/kicad/gen_head_sch.py; edit this file from now on

Design note: docs/head-pcb.md

File: head.kicad_sch

Title: Dukki head carrier

Size: A2 | Date: 16/10/2026 | Rev: B

KiCad E.D.A. 10.0.0 | Id: 1/1